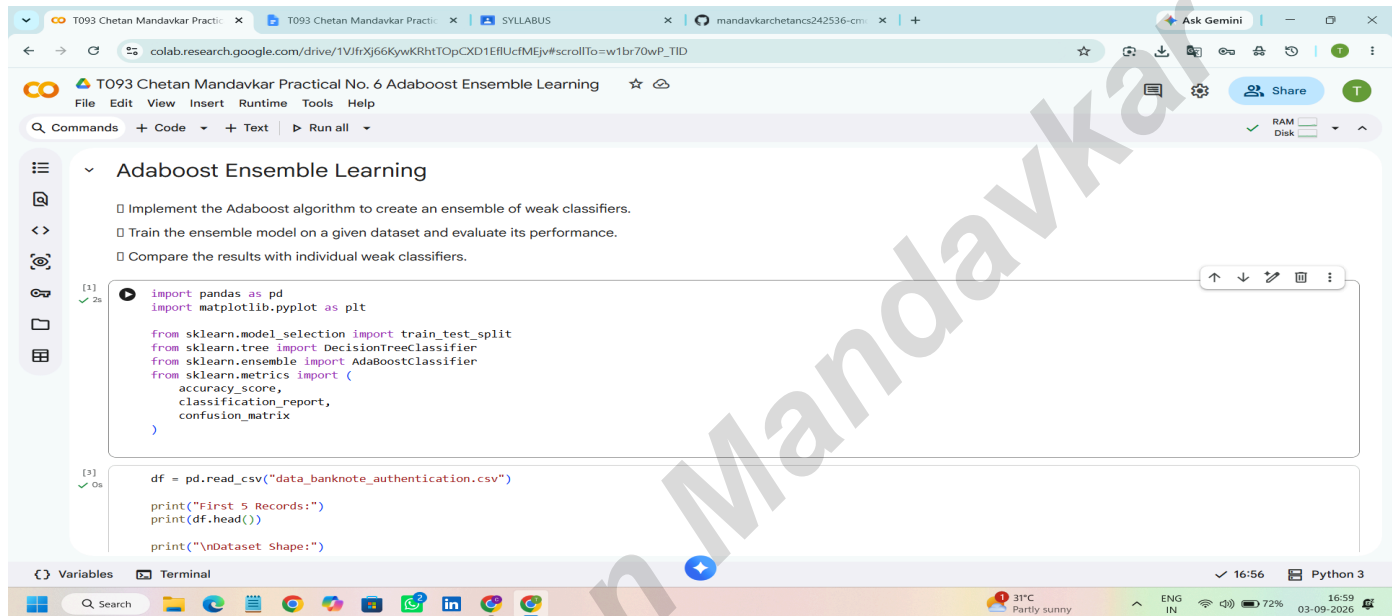


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Aim:- Adaboost Ensemble Learning

Implement the Adaboost algorithm to create an ensemble of weak classifiers. Train the ensemble model on a given dataset and evaluate its performance. Compare the results with individual weak classifiers.



The screenshot shows a Google Colab notebook titled "T093 Chetan Mandavkar Practical No. 6 Adaboost Ensemble Learning". The code is written in Python and includes the following:

```
[1] import pandas as pd
import matplotlib.pyplot as plt

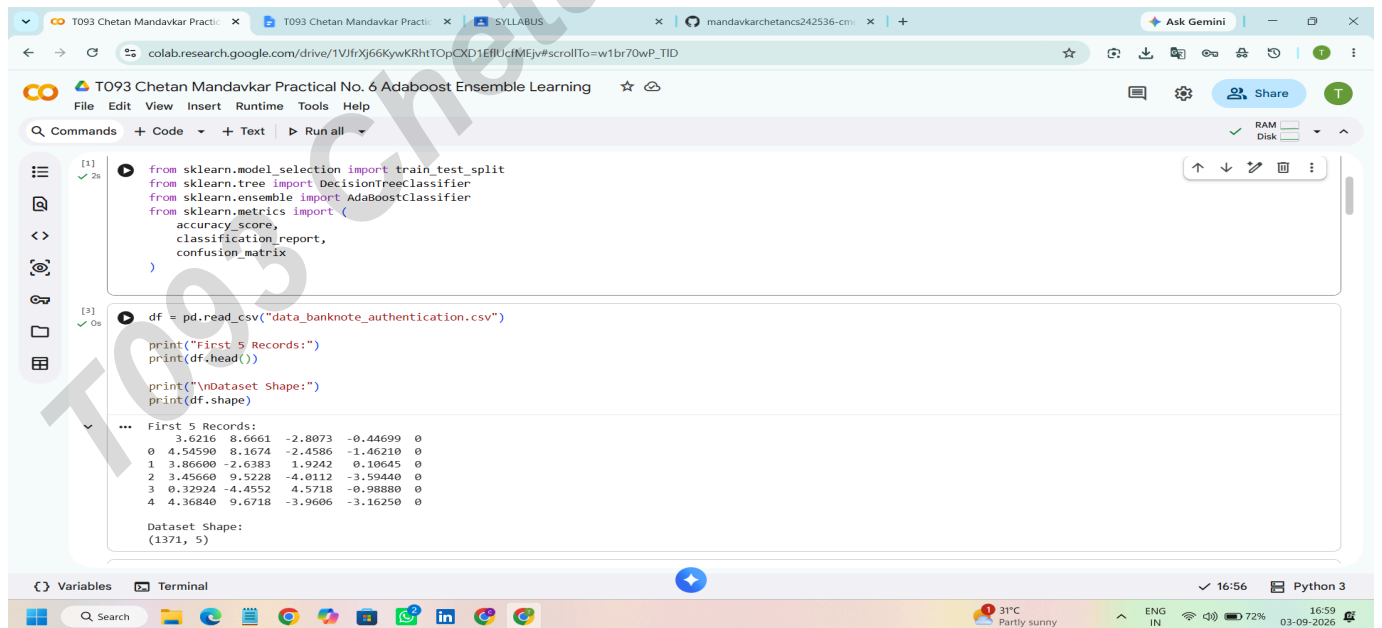
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import AdaBoostClassifier
from sklearn.metrics import (
    accuracy_score,
    classification_report,
    confusion_matrix
)
```

```
[3] df = pd.read_csv("data_banknote_authentication.csv")

print("First 5 Records:")
print(df.head())

print("\nDataset Shape:")
```

The interface includes a left sidebar with icons for file management, a top toolbar with "Commands", "Code", "Text", and "Run all" buttons, and a bottom status bar showing "Python 3" and system information.



The screenshot shows the same Google Colab notebook, but now with the output of the code execution. The output includes the first 5 records of the dataset and the dataset shape.

```
[2] from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import AdaBoostClassifier
from sklearn.metrics import (
    accuracy_score,
    classification_report,
    confusion_matrix
)
```

```
[3] df = pd.read_csv("data_banknote_authentication.csv")

print("First 5 Records:")
print(df.head())

print("\nDataset Shape:")
print(df.shape)
```

```
First 5 Records:
   3.6216  8.6661 -2.8073 -0.44699  0
0  4.54590  8.1674 -2.4586 -1.46210  0
1  3.86600 -2.6383  1.9242  0.10645  0
2  3.45660  9.5228 -4.0112 -3.59440  0
3  0.32924 -4.4552  4.5718 -0.98880  0
4  4.36840  9.6718 -3.9606 -3.16250  0

Dataset Shape:
(1371, 5)
```

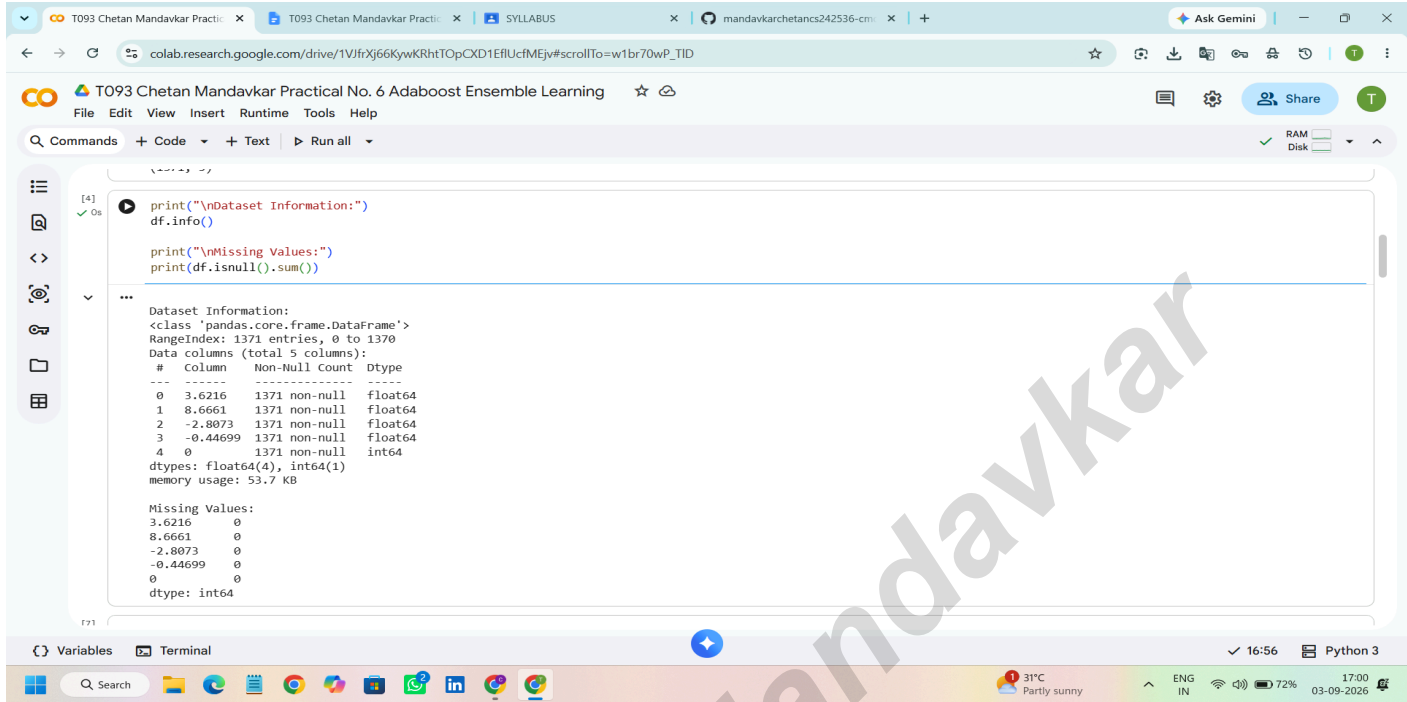
The output is displayed in a scrollable area, and the bottom status bar remains the same.

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The screenshot shows a Google Colab notebook interface. The title bar indicates the notebook is titled "T093 Chetan Mandavkar Practical No. 6 Adaboost Ensemble Learning". The code editor displays the following Python code:

```
[4]: print("\nDataset Information:")
df.info()

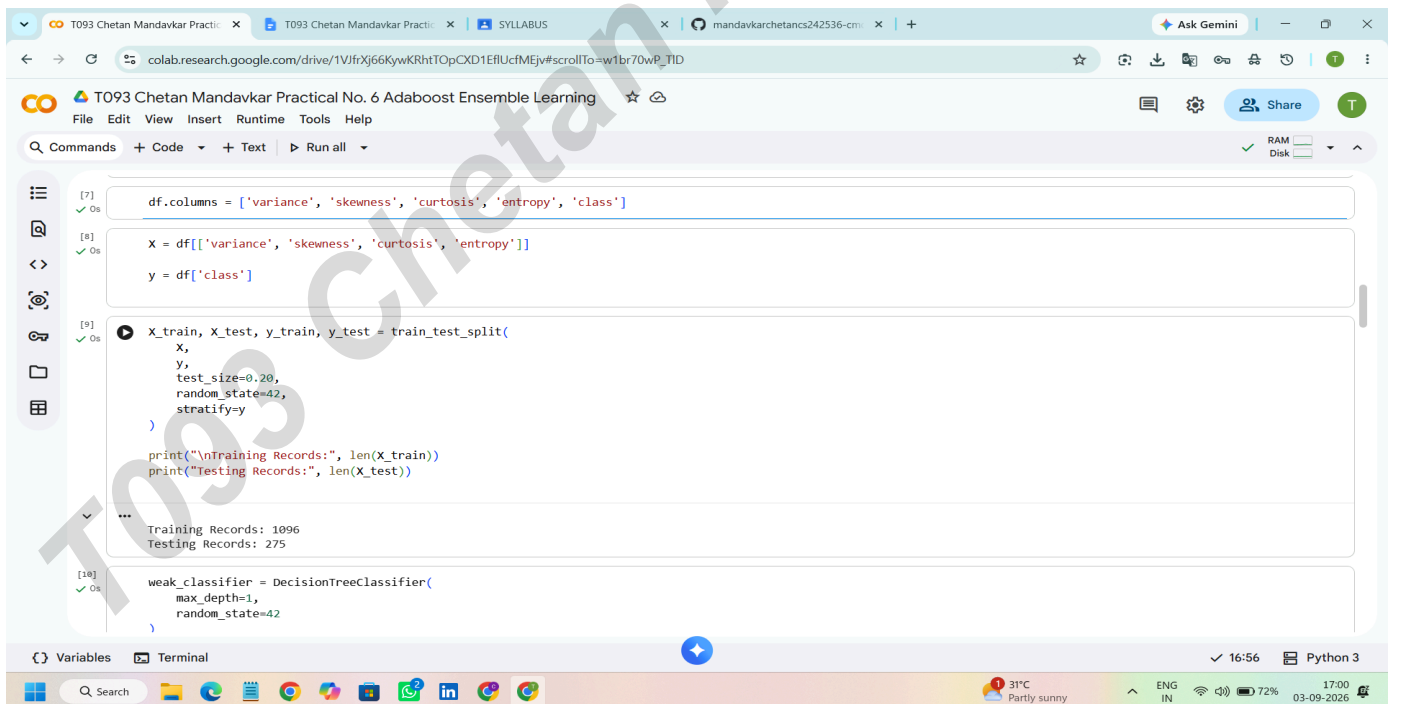
print("\nMissing Values:")
print(df.isnull().sum())
```

The output of the code is displayed below the code cell:

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1371 entries, 0 to 1370
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype  
---  -
 0   3.6216      1371 non-null   float64
 1   8.6661      1371 non-null   float64
 2   -2.8073     1371 non-null   float64
 3   -0.44699    1371 non-null   float64
 4   0           1371 non-null   int64   
dtypes: float64(4), int64(1)
memory usage: 53.7 KB

Missing Values:
3.6216      0
8.6661      0
-2.8073     0
-0.44699    0
0           0
dtype: int64
```

The bottom of the screenshot shows the Windows taskbar with various application icons and system status information, including the time 16:56 and date 03-09-2026.



The screenshot shows a Google Colab notebook interface. The title bar indicates the notebook is titled "T093 Chetan Mandavkar Practical No. 6 Adaboost Ensemble Learning". The code editor displays the following Python code:

```
[7]: df.columns = ['variance', 'skewness', 'curtosis', 'entropy', 'class']

[8]: X = df[['variance', 'skewness', 'curtosis', 'entropy']]
     y = df['class']

[9]: X_train, X_test, y_train, y_test = train_test_split(
     X,
     y,
     test_size=0.20,
     random_state=42,
     stratify=y
)

print("\nTraining Records:", len(X_train))
print("\nTesting Records:", len(X_test))

[10]: weak_classifier = DecisionTreeClassifier(
      max_depth=1,
      random_state=42
    )
```

The output of the code is displayed below the code cell:

```
Training Records: 1096
Testing Records: 275
```

The bottom of the screenshot shows the Windows taskbar with various application icons and system status information, including the time 16:56 and date 03-09-2026.

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The screenshot shows a Google Colab environment with the title "T093 Chetan Mandavkar Practical No. 6 Adaboost Ensemble Learning". The code in the cell [10] defines a `DecisionTreeClassifier` with `max_depth=1` and `random_state=42`. It fits the classifier on `X_train` and `y_train`, then predicts on `X_test`. The accuracy is calculated using `accuracy_score` and printed. The output shows the accuracy of the individual weak classifier is 85.81818181818181 %.

```
[10] ✓ Os
weak_classifier = DecisionTreeClassifier(
    max_depth=1,
    random_state=42
)

weak_classifier.fit(X_train, y_train)

weak_pred = weak_classifier.predict(X_test)

weak_accuracy = accuracy_score(
    y_test,
    weak_pred
)

print("\n-----")
print("INDIVIDUAL WEAK CLASSIFIER")
print("-----")

print("Accuracy:",
      weak_accuracy * 100, "%")

...

INDIVIDUAL WEAK CLASSIFIER
-----
Accuracy: 85.81818181818181 %
```

The screenshot shows the continuation of the Google Colab environment. The code in cell [11] creates an `AdaBoostClassifier` using the `DecisionTreeClassifier` as the estimator, with `n_estimators=50` and `learning_rate=1.0`. Cell [12] fits the AdaBoost model on the training data and prints "AdaBoost Training Completed!". Cell [13] uses the trained model to predict on the test data. Cell [14] calculates the accuracy of the AdaBoost ensemble and prints it. The output shows the accuracy of the AdaBoost ensemble is 85.81818181818181 %.

```
[11] ✓ Os
ada_model = AdaBoostClassifier(
    estimator=DecisionTreeClassifier(
        max_depth=1,
        random_state=42
    ),
    n_estimators=50,
    learning_rate=1.0,
    random_state=42
)

[12] ✓ Os
ada_model.fit(X_train, y_train)

print("\nAdaBoost Training Completed!")

...
AdaBoost Training Completed!

[13] ✓ Os
ada_pred = ada_model.predict(X_test)

[14] ✓ Os
ada_accuracy = accuracy_score(
    y_test,
    ada_pred
)

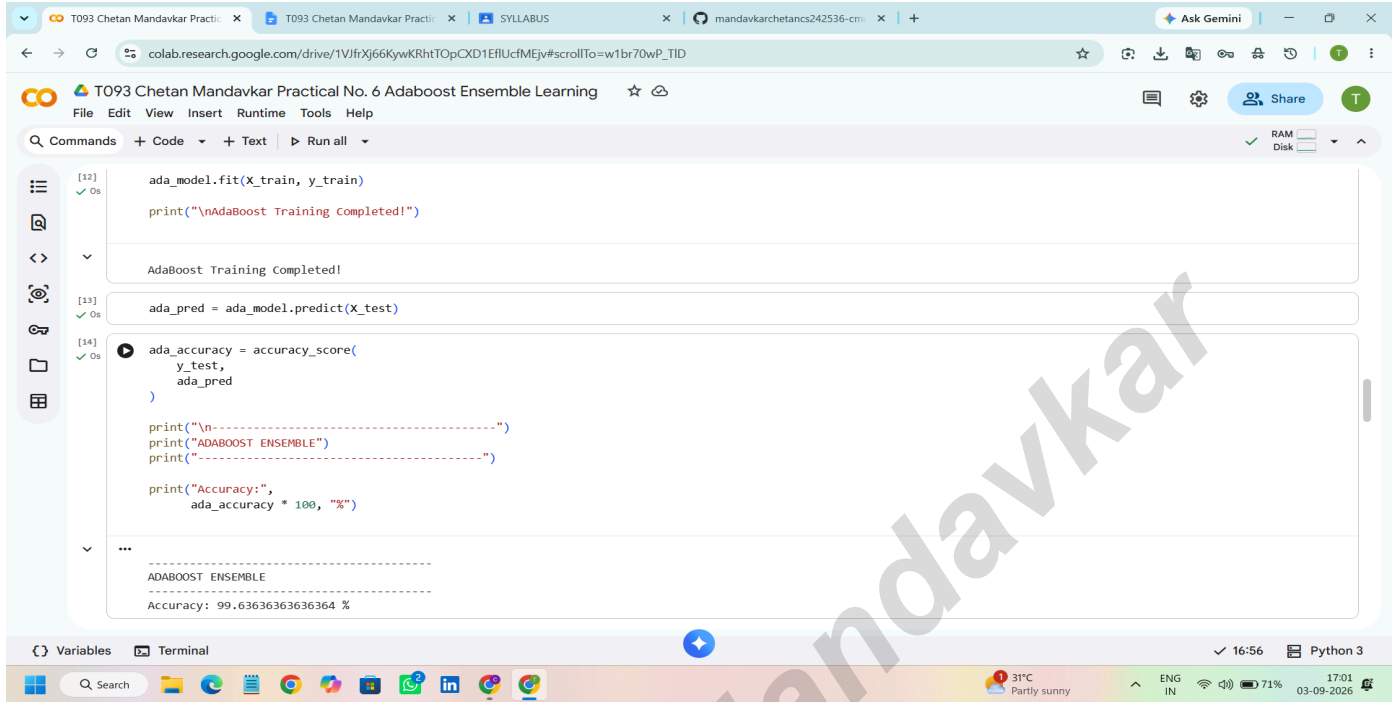
print("\n-----")
print("ADABOOST ENSEMBLE")

...

ADABOOST ENSEMBLE
-----
Accuracy: 85.81818181818181 %
```

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The screenshot shows a Google Colab notebook titled "T093 Chetan Mandavkar Practical No. 6 Adaboost Ensemble Learning". The notebook is open to a cell containing Python code for training an AdaBoost model and calculating its accuracy. The code is as follows:

```
[12] ada_model.fit(X_train, y_train)
      print("\nAdaBoost Training Completed!")

AdaBoost Training Completed!

[13] ada_pred = ada_model.predict(X_test)

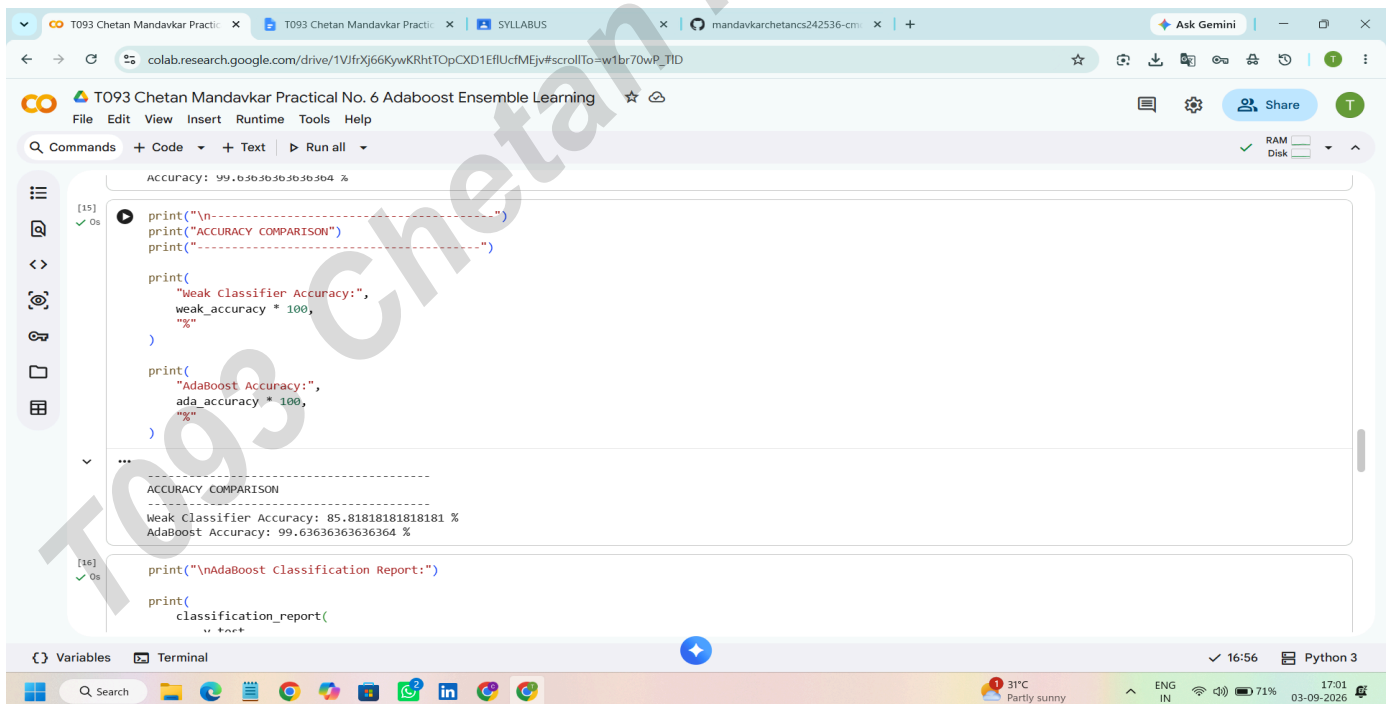
[14] ada_accuracy = accuracy_score(
      y_test,
      ada_pred
    )

      print("\n-----")
      print("ADABOOST ENSEMBLE")
      print("-----")

      print("Accuracy:",
            ada_accuracy * 100, "%")

-----
ADABOOST ENSEMBLE
-----
Accuracy: 99.636363636364 %
```

The bottom of the notebook shows the "Variables" tab with a list of variables and the "Terminal" tab. The system tray at the bottom indicates the time is 16:56 and the date is 03-09-2026.



The screenshot shows the same Google Colab notebook, now with additional code for comparing the accuracy of the weak classifier and the AdaBoost ensemble, and generating a classification report. The code is as follows:

```
Accuracy: 99.636363636364 %

[15] print("\n-----")
      print("ACCURACY COMPARISON")
      print("-----")

      print(
        "Weak Classifier Accuracy:",
        weak_accuracy * 100,
        "%"
      )

      print(
        "AdaBoost Accuracy:",
        ada_accuracy * 100,
        "%"
      )

-----
ACCURACY COMPARISON
-----
Weak Classifier Accuracy: 85.818181818181 %
AdaBoost Accuracy: 99.636363636364 %

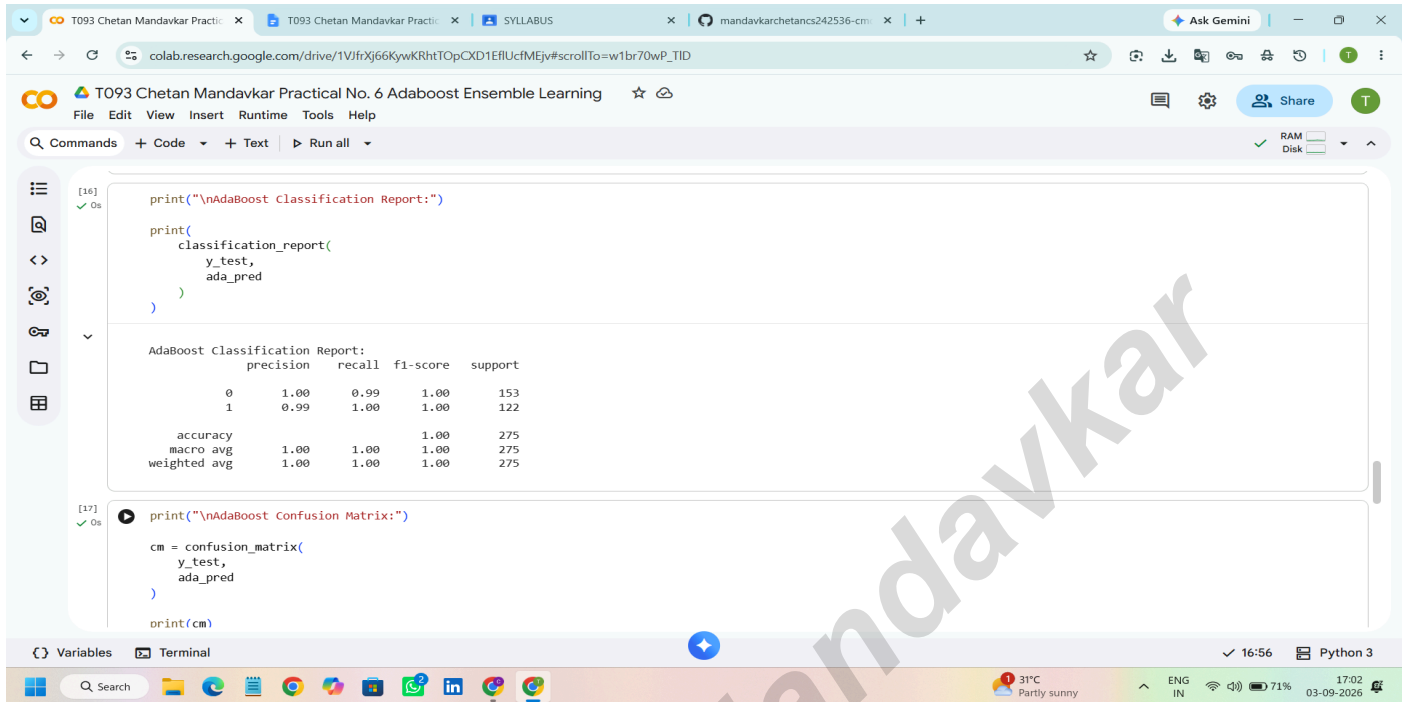
[16] print("\nAdaBoost Classification Report:")

      print(
        classification_report(
          y_test,
```

The bottom of the notebook shows the "Variables" tab with a list of variables and the "Terminal" tab. The system tray at the bottom indicates the time is 16:56 and the date is 03-09-2026.

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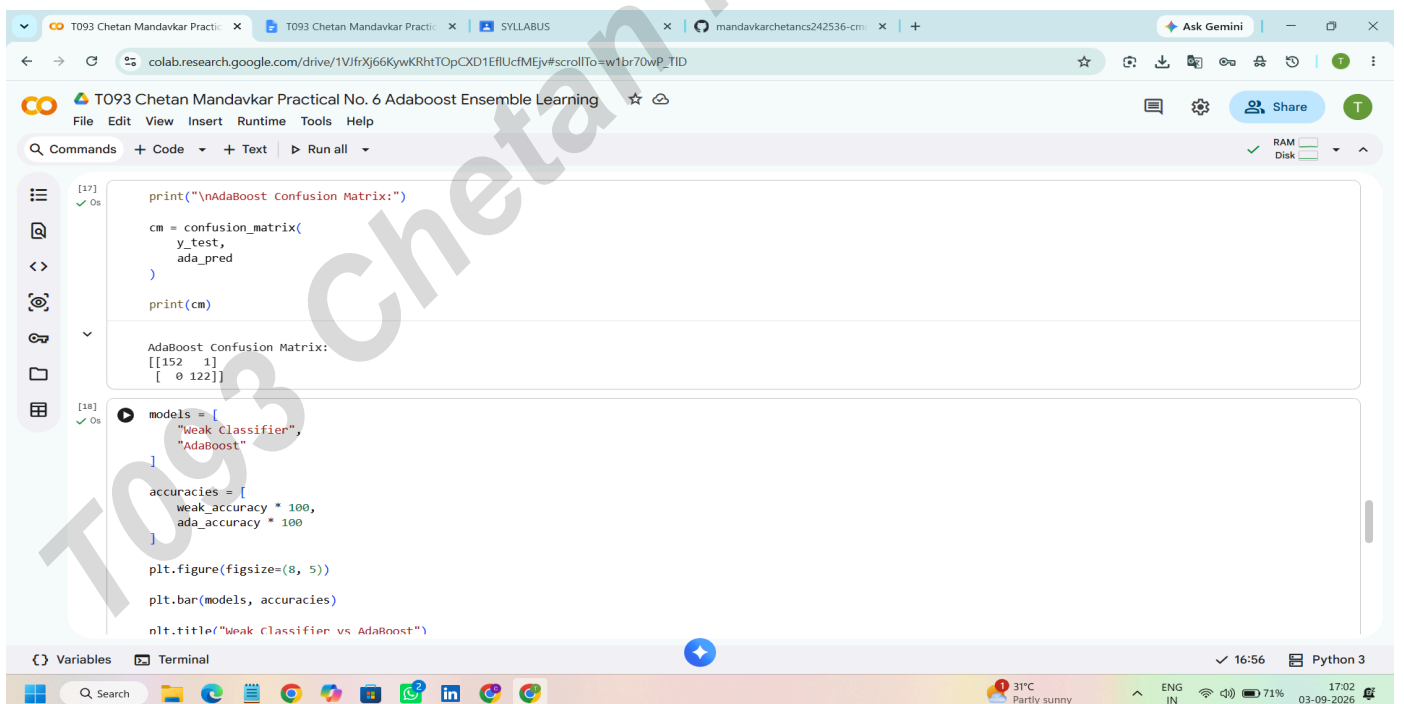
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The screenshot shows a Google Colab environment with the title "T093 Chetan Mandavkar Practical No. 6 Adaboost Ensemble Learning". The code cell [16] contains a call to `classification_report` with `y_test` and `ada_pred` as arguments. The output displays the "AdaBoost Classification Report" with the following data:

	precision	recall	f1-score	support
0	1.00	0.99	1.00	153
1	0.99	1.00	1.00	122
accuracy			1.00	275
macro avg	1.00	1.00	1.00	275
weighted avg	1.00	1.00	1.00	275

Below this, cell [17] contains code to generate a confusion matrix using `confusion_matrix` and `print(cm)`.



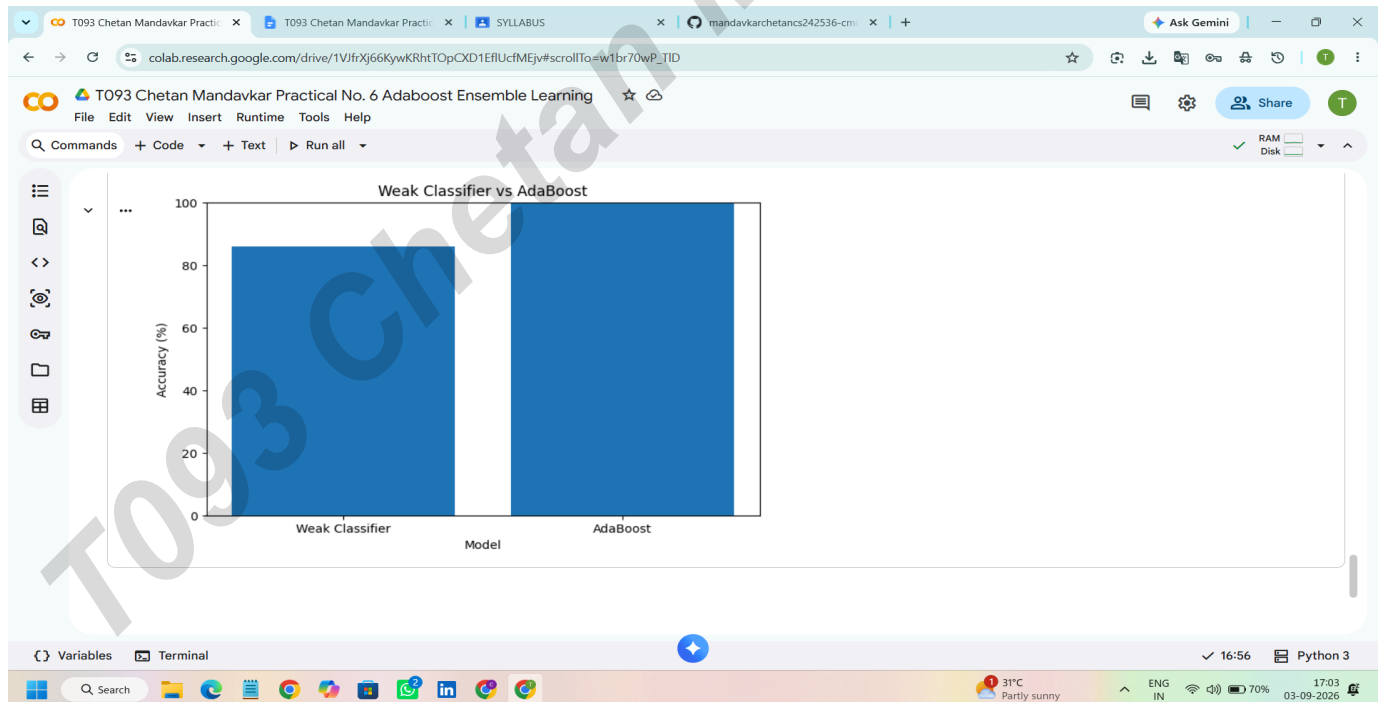
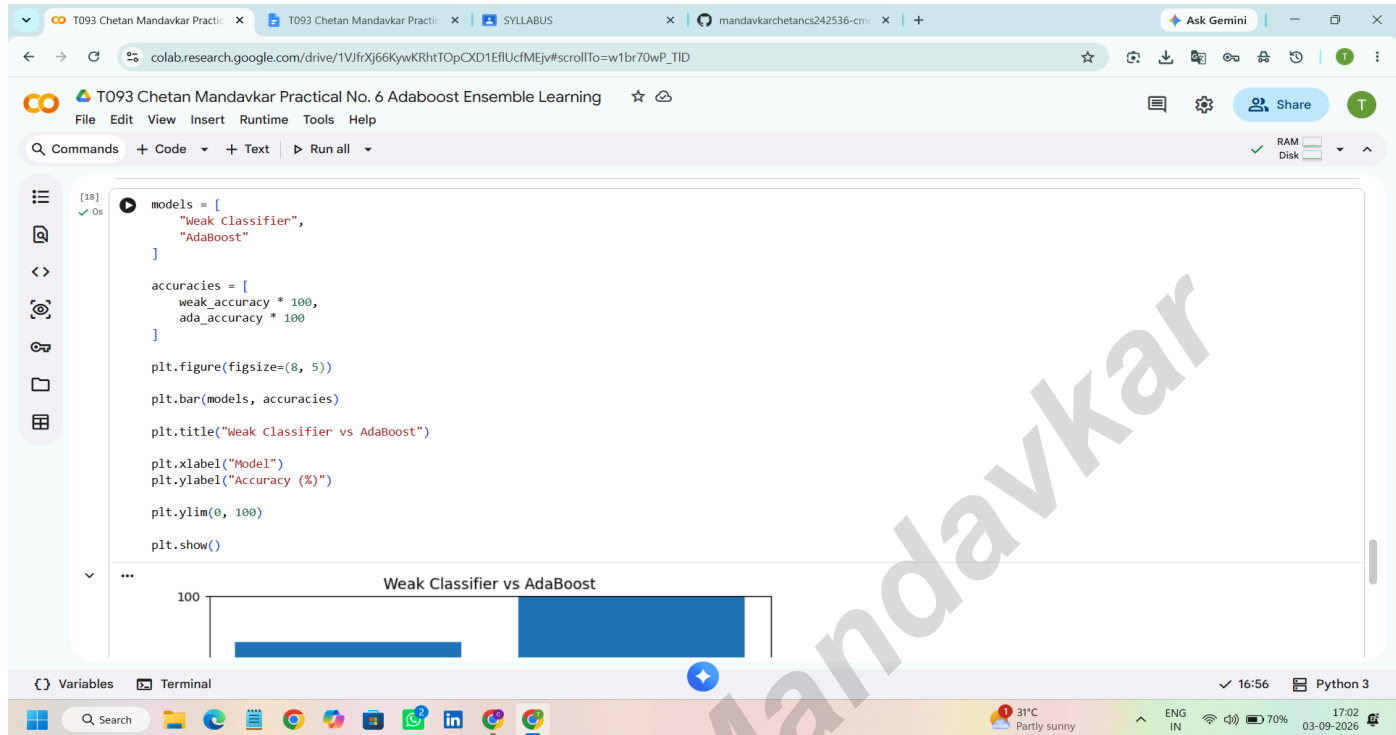
The screenshot shows the continuation of the Google Colab environment. Cell [17] displays the "AdaBoost Confusion Matrix" as a 2x2 array: `[[152 1], [ 0 122]]`. Cell [18] contains code to compare the accuracies of a "Weak Classifier" and "AdaBoost". The output shows the following data:

Model	Accuracy
Weak Classifier	100
AdaBoost	100

The code also includes `plt.figure(figsize=(8, 5))`, `plt.bar(models, accuracies)`, and `plt.title("Weak Classifier vs AdaBoost")` to visualize the comparison.

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